Light

GCSE Computer Science - Network Protocols (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which connection mode uses Ethernet cables to connect devices to a network?

- A. Wireless connection
- B. Bluetooth connection
- C. Wired connection
- D. Satellite connection

Q2 (1 mark)

What does Wi-Fi use to transmit data between devices?

- A. Ethernet cables

- B. Fibre optic cables
- C. Radio waves
- D. Infrared light

Q3 (1 mark)

What is Bluetooth primarily designed for?

- A. Short-range wireless communication between devices
- B. Connecting to the internet across large distances
- C. Replacing all wired network connections
- D. Providing faster speeds than Wi-Fi

Q4 (1 mark)

What format does a MAC address use?

- A. Six pairs of hexadecimal digits, often separated by colons
- B. Four numbers separated by dots, each 0-255
- C. A single large decimal number
- D. Eight groups of four hexadecimal digits separated by colons

Q5 (1 mark)

Which of these is a valid IPv4 address format?

- A. 192.168.1.1
- B. 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- C. AA:BB:CC:DD:EE:FF
- D. www.example.com

Q6 (1 mark)

How many bits does an IPv6 address contain?

- A. 32 bits
- B. 48 bits
- C. 64 bits
- D. 128 bits

Q7 (1 mark)

In the TCP/IP model, which layer is responsible for reliable data delivery between applications?

- A. Transport layer
- B. Application layer
- C. Internet layer
- D. Network Access layer

Q8 (1 mark)

What is the main function of the Internet layer in the TCP/IP model?

- ○ **A.** Addressing and routing packets across networks
- ○ **B.** Connecting cables to devices
- ○ **C.** Displaying web pages to users
- ○ **D.** Encrypting all network data

Q9 (1 mark)

Which TCP/IP layer handles the physical transmission of data over network hardware?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Network Access layer

Q10 (1 mark)

What is the difference between a protocol and a standard?

- ○ **A.** A protocol defines communication rules; a standard ensures products from different manufacturers work together
- ○ **B.** They are exactly the same thing
- ○ **C.** A standard is software; a protocol is hardware

- ○ **D.** Protocols are for wired networks; standards are for wireless networks

Q11 (1 mark)

Which port number is typically used for HTTP traffic?

- ○ **A.** Port 21
- ○ **B.** Port 25
- ○ **C.** Port 80
- ○ **D.** Port 443

Q12 (1 mark)

Which port number is typically used for HTTPS (secure web) traffic?

- ○ **A.** Port 80
- ○ **B.** Port 110
- ○ **C.** Port 443
- ○ **D.** Port 8080

Q13 (2 marks)

A school is deciding between wired Ethernet and Wi-Fi for their new computer suite.
Which statement correctly compares these connection types?

- ○ **A.** Wi-Fi is always faster and more reliable than Ethernet
- ○ **B.** Ethernet provides faster, more reliable connections but requires physical cabling; Wi-Fi offers flexibility but may suffer from interference
- ○ **C.** Ethernet and Wi-Fi have identical performance in all situations
- ○ **D.** Wi-Fi can only be used outdoors, while Ethernet is for indoor use only

Q14 (2 marks)

A user wants to connect wireless headphones to their laptop for a video call.
Why is Bluetooth an appropriate choice for this scenario?

- ○ **A.** Bluetooth provides the fastest possible connection speed
- ○ **B.** Bluetooth requires no pairing process between devices
- ○ **C.** Bluetooth can connect devices over distances of several kilometres
- ○ **D.** Bluetooth is designed for short-range device connections and uses minimal power, ideal for personal peripherals like headphones

Q15 (2 marks)

Why was IPv6 developed to replace IPv4?

- ○ **A.** IPv4 was too secure and needed to be simplified
- ○ **B.** IPv4's 32-bit address space was running out of unique addresses as more devices connected to the internet
- ○ **C.** IPv4 was too slow for modern networks

○ **D.** IPv6 was created solely to support wireless networks

Q16 (2 marks)

What is a key difference between MAC addresses and IP addresses?

○ **A.** MAC addresses can be changed easily; IP addresses are permanently fixed to hardware

○ **B.** IP addresses are used within a local network; MAC addresses are used across the internet

○ **C.** MAC addresses are longer than IP addresses in all cases

○ **D.** MAC addresses are assigned by manufacturers and are hardware-based; IP addresses are logical and can be assigned by the network

Q17 (2 marks)

When data is sent from one computer to another, it passes through the TCP/IP layers.
In what order do the layers process data when sending?

○ **A.** Network Access → Internet → Transport → Application

○ **B.** Internet → Application → Network Access → Transport

○ **C.** Transport → Network Access → Application → Internet

○ **D.** Application → Transport → Internet → Network Access

Q18 (2 marks)

An office has desktop computers that rarely move and require consistent, high-speed network access for large file transfers.
Which connection type would be most appropriate?

- ○ **A.** Bluetooth because it requires no cables
- ○ **B.** Wi-Fi because it is the newest technology
- ○ **C.** Wired Ethernet because it provides consistent, high-speed connections for stationary devices
- ○ **D.** Satellite because it covers large areas

Q19 (2 marks)

When a browser connects to a website using HTTPS, what happens before data is exchanged?

- ○ **A.** The browser downloads a password from the server
- ○ **B.** A handshake process establishes encryption keys so data can be transmitted securely
- ○ **C.** The server sends all data in plain text first
- ○ **D.** The user must manually enter an encryption key

Q20 (2 marks)

A user configures their email application with POP3 settings. They download emails on their laptop, then later check email on their phone and find the inbox empty.
What explains this behaviour?

- ○ **A.** The phone has a faulty email application
- ○ **B.** The emails were automatically deleted after 24 hours

- ○ **C.** POP3 only works with laptops, not phones
- ○ **D.** POP3 downloads emails and typically removes them from the server, so they are no longer available for other devices

Q21 (2 marks)

A network technician cannot access any websites but can successfully ping the web server's IP address.
Which layer should they investigate first?

- ○ **A.** Network Access layer because the cables must be faulty
- ○ **B.** Internet layer because the IP address is not working
- ○ **C.** Application layer because HTTP/HTTPS is not functioning even though lower layers work
- ○ **D.** Transport layer because TCP is always the problem

Q22 (2 marks)

FTP uses two port numbers: one for commands and one for data transfer.
What are the standard port numbers for FTP?

- ○ **A.** Port 80 for commands, port 443 for data
- ○ **B.** Port 25 for commands, port 110 for data
- ○ **C.** Port 443 for commands, port 80 for data
- ○ **D.** Port 21 for commands, port 20 for data

Q23 (2 marks)

Organisations like IEEE (Institute of Electrical and Electronics Engineers) create networking standards.
Why is their role important?

- ○ **A.** They manufacture all networking equipment
- ○ **B.** They create specifications that ensure devices from different manufacturers can communicate and work together
- ○ **C.** They control the internet and can shut it down
- ○ **D.** They provide free internet access worldwide

Q24 (3 marks)

A user sends an email using their email application. Describe how the email data passes through the TCP/IP layers on the sender's computer, identifying what happens at each layer.

- ○ **A.** Application: email is created → Transport: split into packets with port numbers → Internet: IP addresses added for routing → Network Access: converted to signals for transmission
- ○ **B.** Network Access: signals received → Internet: IP checked → Transport: packets combined → Application: email displayed
- ○ **C.** All layers process the email simultaneously with no specific order
- ○ **D.** Only the Application layer is involved in sending emails

Q25 (3 marks)

A factory needs networking for three scenarios: fixed machinery on the production floor, tablets used by supervisors walking around, and a sensor that transmits temperature data to a nearby display unit.

Analyse which connection type suits each scenario.

- ○ **A.** Wi-Fi for everything because it is the most modern technology
- ○ **B.** Bluetooth for everything because it uses the least power
- ○ **C.** Satellite connections for all scenarios to ensure maximum coverage
- ○ **D.** Wired Ethernet for fixed machinery (reliability), Wi-Fi for mobile tablets (flexibility), and Bluetooth for the sensor to nearby display (short-range, low-power)

Q26 (3 marks)

A new business is setting up their email system. Employees will send emails to clients and colleagues, receive emails on multiple devices (phone, laptop, desktop), and need emails to remain synchronised across all devices.
Analyse which protocols are needed and why.

- ○ **A.** HTTP for sending and FTP for receiving emails on multiple devices
- ○ **B.** SMTP for sending emails; IMAP for receiving emails because it keeps emails on the server, enabling synchronisation across multiple devices
- ○ **C.** POP3 for both sending and receiving to ensure local storage
- ○ **D.** HTTPS for all email operations because it is secure

Q27 (3 marks)

A company is evaluating the security of different protocols used in their network.
Analyse which of these scenarios presents the greatest security risk.

- ○ **A.** Using HTTPS for all website access
- ○ **B.** Using FTP to transfer sensitive files because login credentials and file contents are transmitted in plain text without encryption
- ○ **C.** Using IMAP with TLS encryption for email
- ○ **D.** Using Bluetooth with PIN pairing for wireless peripherals

Q28 (3 marks)

A medium-sized company has 500 employees who need IP addresses. They are considering IPv4 with a private address range.
Analyse why using private IPv4 addresses with Network Address Translation (NAT) is appropriate.

- ○ **A.** Private addresses are faster than public addresses
- ○ **B.** Private addresses cannot connect to the internet at all
- ○ **C.** Private addresses allow many internal devices to share limited public IP addresses through NAT, conserving the scarce IPv4 address space while maintaining internet connectivity
- ○ **D.** Private addresses eliminate the need for any network security

Q29 (3 marks)

A software company is developing a new application that needs to send data across the internet.
Analyse how the layered TCP/IP model benefits their development process.

- ○ **A.** Developers must understand and implement all four layers themselves
- ○ **B.** The layered model means developers only need to work with the Application layer; the operating system and network hardware handle

lower layers automatically

- ○ **C.** The layers are purely theoretical and provide no practical benefit
- ○ **D.** Each layer requires a separate programming language

Q30 (3 marks)

A video streaming service needs to deliver live video to users. Occasionally, a few video frames may be lost during transmission. Analyse why the service might choose UDP (User Datagram Protocol) instead of TCP for streaming.

- ○ **A.** UDP is more secure than TCP for video streaming
- ○ **B.** UDP provides faster transmission without the overhead of acknowledgements and retransmission; for live video, a few lost frames are acceptable because retransmitting old frames would cause more disruption than skipping them
- ○ **C.** UDP guarantees that every frame arrives correctly
- ○ **D.** TCP cannot be used for any video-related applications

Total: **55** marks • Time: **30 min**

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